Exploring how smoking status depends on other variables

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1 Introduction

This report will investigate how smoking status depends on other variables ranging from a measure of sex, age, race and other social factors. This data is from a sample in the US, and this report aims to find an apt model for smoking status in relation to specified variables.

Summary	
Variable	**Description**
Smoker	1 if current smoker, 0 otherwise
Ban	1 if there is a work area ban on smoking, 0 otherwise
Age	Age of the individuals in years
Age2	Age of the individual squared
Education	Measure of the highest level of education: 1 is high school (hs) dropout, 2 hs graduate, 3 some college, 4 college graduate, 5 master's degree or higher
African- American or Hispanic (Aahisp)	1 if they are African-American or Hispanic, 0 otherwise
Female	1 if female, 0 otherwise

2 Exploratory Data Analysis

We are exploring how each of these variables potentially impact the smoking status. Initially let's note that there's more non-smokers than active smokers (if you take the mean value it's 0.2421 which is closer to 0 than 1). Similar summaries can be deduced for the other factors: ban mean is 0.6098 (most work places ban smoking), Aahisp mean is 0.1872 (the majority of the data set are people not identifying as Aahisp), and female mean is 0.5637 (there's marginally more females than men). The median highest education level is attending some college and 50 percent of people fall between being a high school graduate and a college graduate. These simple summaries help us understand the structure of the data. Now looking into the plots more closely:

Figure 1's histogram of age appears to be approximately normal, with a possible peak in younger working-age adults. Age is therefore likely to have a non-linear relationship with smoking by comparing that with higher ages. This therefore motivates and justifies the inclusion of age^2 as one of our parameters for the model.

Figure 2 shows that the bar chart of education levels indicates a higher representation of individuals with "Some College" or "High School Graduate" levels. This variable will likely have a significant impact on smoking behaviour.

Figure 3 (with the aid of Figure 2) is perhaps the most interesting; the proportion bar chart reveals a 'negative' relationship between education level and smoking prevalence, with higher education levels corresponding to lower smoking rates.

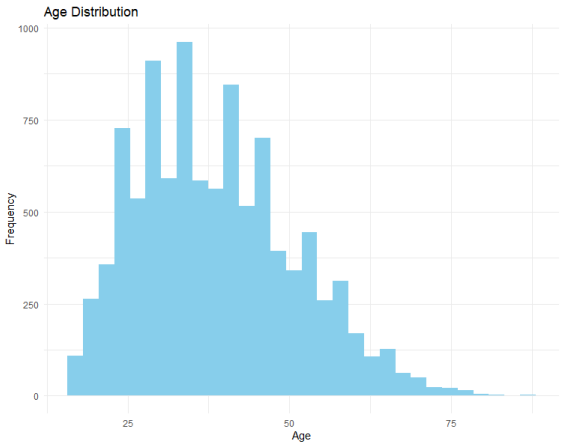


Figure 1: Univariate plot for the age distribution

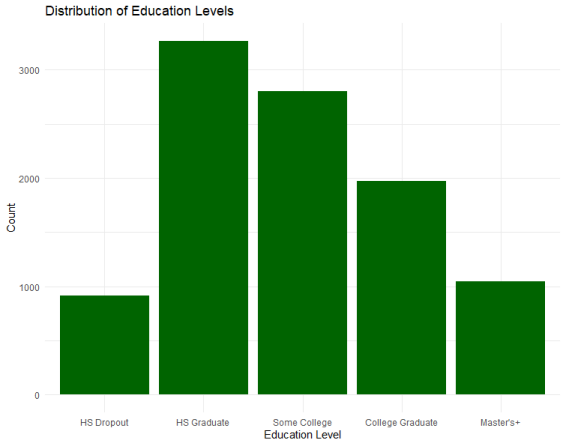


Figure 2: Univariate plot for the education levels to understand the education level composition which is a key variable that we will explore later

Figure 4 This bar chart suggests that workplace smoking bans are associated with a lower prevalence of smoking as there's a difference of about 7.75 percent between the proportion of smokers, indicating a potential protective effect of having a workplace ban.

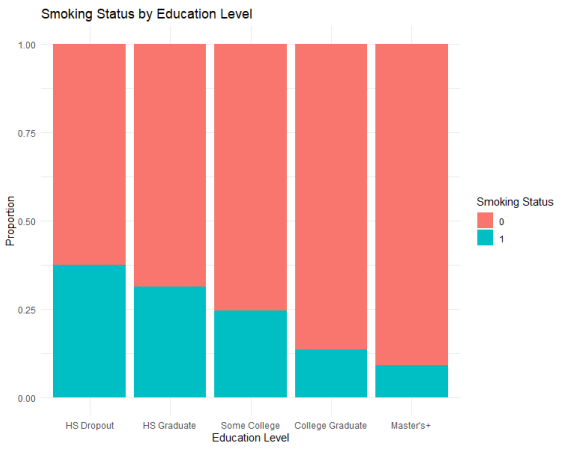


Figure 3: Bivariate plot showing how smoking prevalence changes with education level

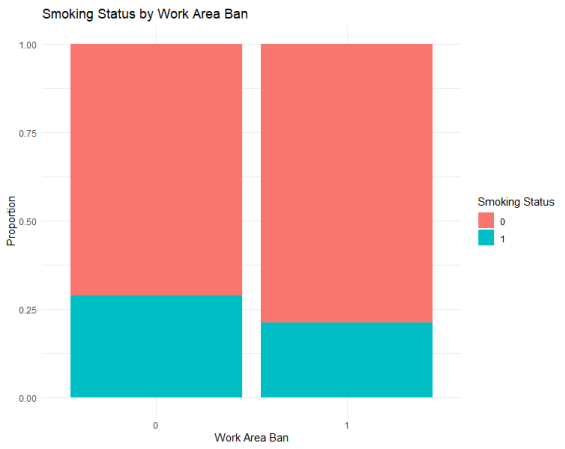


Figure 4: Bivariate plot to see the influence of workplace smoking bans on smoking status

Figure 5 chart highlights differences in smoking rates by racial/ethnic group, which could reflect socioeconomic or cultural factors influencing smoking behaviour. However this difference in proportion is only about 1.92 percent indicating that potentially people that identify as Aaship smoke less.

Figure 6 bar chart shows whether smoking prevalence differs significantly between males and females. The proportion difference is only about 2.75 percent indicating that females marginally smoke less than males.

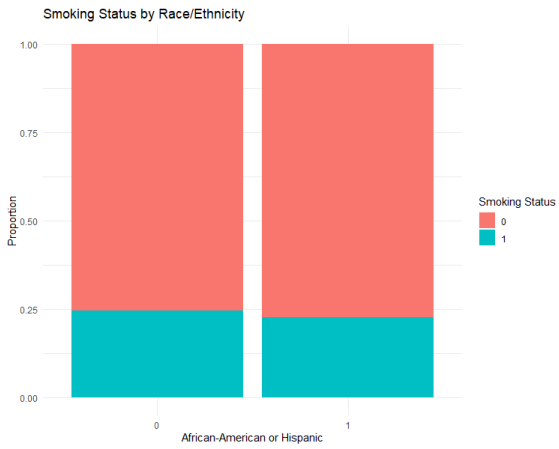


Figure 5: Bivariate plot to explore any disparities in smoking rates across racial/ethnic groups

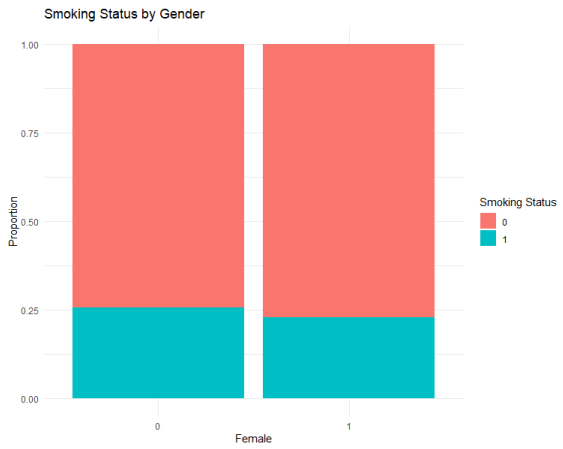


Figure 6: Bivariate plot to identify potential sex differences in smoking prevalence

3 Modelling

3.1 Model

We start by deciding what General Linear Model we want to fit from the exponential family of distributions. Firstly we note that we are only considering interactions with aahisp as that was specified to us. The one that fits best is Binomial with logit (odds ratio) canonical link. This is mainly due to two reasons:

- The dependent variable (smoking status) is a binary term, and the binomial is designed for this
- Canonical link function logit is used as the logit uses the log-odds. This means the coefficients in a binomial with logit link can be interpreted in terms of odds ratios, providing a meaningful way to understand the impact of the explanatory variables on smoking. Above all, this aligns with our objective hence it's the best choice

```
    smoke_glm <- glm(smoker ~ (aahisp * (ban + age + age2 + education + female)),
        family = binomial(link = "logit"), data = smoke)
```

3.2 Model tests and revisions

Now that we've chosen our GLM, we need to run a few tests to determine the importance and impact of certain variables. To do this we will start with a Likelihood Ratio Test (LRT) for two nested models using the Deviance measure. We will test all possible models by looking at what happens when a parameter is removed and comparing that to the output of a chi-squared distribution. We will define a significance level of 0.05. Firstly we should highlight the candidates for removal which we can see in the ANOVA table at the last column, and see which have a $p > 0.05$ Initially the full model that we started with is:

```
    smoke_glm <- glm(smoker ~ (aahisp * (ban + age + age2 + education + female)),
        family = binomial(link = "logit"), data = smoke)
    Call:
glm(formula = smoker ~ (aahisp * (ban + age + age2 + education +
    female)), family = binomial(link = "logit"), data = smoke)

Coefficients:
                              Estimate Std. Error z value Pr(>|z|)
(Intercept)                 -0.7014285  0.2696275  -2.601  0.00928 **
aahisp                      -1.5524791  0.6271478  -2.475  0.01331 *
ban                         -0.2783246  0.0547419  -5.084 3.69e-07 ***
age                          0.0544843  0.0132515   4.112 3.93e-05 ***
age2                        -0.0007837  0.0001590  -4.927 8.34e-07 ***
```

```
educationHS Graduate              -0.5715498  0.1035317  -5.521 3.38e-08 ***
educationSome College             -0.9550475  0.1072430  -8.905  < 2e-16 ***
educationCollege Graduate         -1.7720931  0.1191644 -14.871  < 2e-16 ***
educationMasters+                 -2.1715767  0.1470895 -14.764  < 2e-16 ***
female                            -0.1404235  0.0546771  -2.568  0.01022 *
aahisp:ban                         0.1086164  0.1271715   0.854  0.39305
aahisp:age                         0.0258547  0.0320005   0.808  0.41912
aahisp:age2                       -0.0001252  0.0003915  -0.320  0.74912
aahisp:educationHS Graduate        0.3952412  0.1783184   2.216  0.02666 *
aahisp:educationSome College       0.5832370  0.1863782   3.129  0.00175 **
aahisp:educationCollege Graduate   1.0745844  0.2509264   4.282 1.85e-05 ***
aahisp:educationMasters+           0.8691722  0.4405886   1.973  0.04852 *
aahisp:female                     -0.2605090  0.1260156  -2.067  0.03871 *
---
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 11070  on 9999  degrees of freedom
Residual deviance: 10442  on 9982  degrees of freedom
AIC: 10478

Number of Fisher Scoring iterations: 4
```

So now the candidates that we would like to investigate with an LRT are aahisp:ban, aahisp:age, and aahisp:age2. Now we perform an iterative LRT to select a new model. This leaves us with a final model:

```
Call:
glm(formula = smoker ~ aahisp + ban + age + age2 + education +
    female + aahisp:age2 + aahisp:education + aahisp:female,
    family = binomial(link = "logit"), data = smoke)

Coefficients:
                                  Estimate Std. Error z value Pr(>|z|)
(Intercept)                     -7.903e-01  2.495e-01  -3.168 0.001536 **
aahisp                          -1.030e+00  1.863e-01  -5.528 3.23e-08 ***
ban                             -2.587e-01  4.941e-02  -5.235 1.65e-07 ***
age                              5.889e-02  1.205e-02   4.886 1.03e-06 ***
age2                            -8.361e-04  1.453e-04  -5.756 8.64e-09 ***
educationHS Graduate            -5.753e-01  1.035e-01  -5.559 2.71e-08 ***
educationSome College           -9.601e-01  1.072e-01  -8.958  < 2e-16 ***
educationCollege Graduate       -1.781e+00  1.190e-01 -14.965  < 2e-16 ***
educationMasters+               -2.182e+00  1.468e-01 -14.865  < 2e-16 ***
female                          -1.433e-01  5.459e-02  -2.625 0.008665 **
aahisp:age2                      1.877e-04  6.643e-05   2.825 0.004725 **
aahisp:educationHS Graduate      4.093e-01  1.778e-01   2.302 0.021345 *
aahisp:educationSome College     6.086e-01  1.849e-01   3.291 0.000998 ***
aahisp:educationCollege Graduate 1.114e+00  2.488e-01   4.476 7.61e-06 ***
aahisp:educationMasters+         9.225e-01  4.384e-01   2.104 0.035344 *
aahisp:female                   -2.463e-01  1.252e-01  -1.967 0.049130 *
---
(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 11070  on 9999  degrees of freedom
Residual deviance: 10444  on 9984  degrees of freedom
AIC: 10476

Number of Fisher Scoring iterations: 4
```

As a final measure as well, we can calculate the Akaike information criterion (AIC) and see if the this has decreased to indicate an improvement in the model:

- AIC of full model: 10478.41
- AIC of fianl model: 10475.84

Clearly we see and decreased and hence an improvement in the model.

4 Model Diagnostics

4.1 Diagnostic Checks

Before interpreting and analysing the fitted coefficients, we must run some diagnostics to see if there are any issue with the data and model. Here we are looking for influential high leverage points that can skew our data and make our fitted coefficients less reliable meaning the interpretation of the coefficients would be wrong.

We look at the Cook's distance to identify points that would be advised to removed. We look for points with Cook's distance such that:

$$C_k \geq \frac{8}{n-2p}$$

and then remove them to get something of the affect

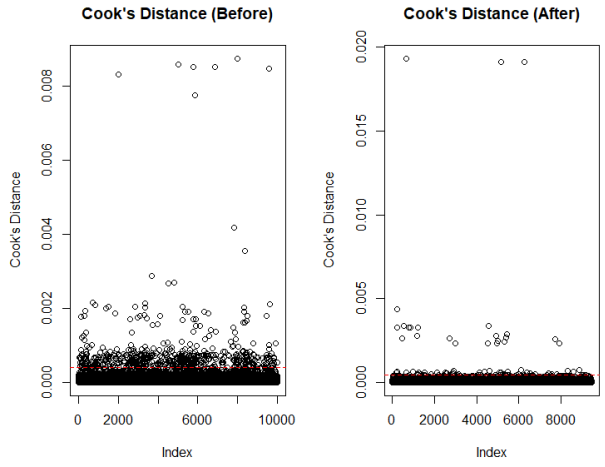


Figure 7: We see that we've removed many points that lie beyond our Cook's distance

And now we would like to see if there has been an improvement by looking at how the leverage of our points changed: We clearly see that there has been an improvement as influential high leverage point have

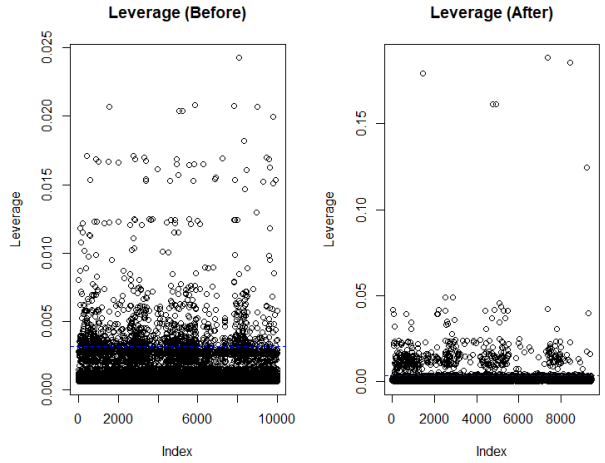


Figure 8: Comparison of the leverage plots

been removed, which would mean that our results would be less skewed and more reliable. In addition,

for this case, looking at deviance residuals and qq-plots aren't exactly useful as they residuals in GLMs are not normally distributed however, it can still give us an insight onto symmetry. Secondarily, in GLMs (particularly binomial) there's heteroscedasticity so vilates the purpose. Furthermore, deviance residuals are often asymmetric for binary data which makes it challenging to assess if the distribution looks 'balanced' or 'normal'.

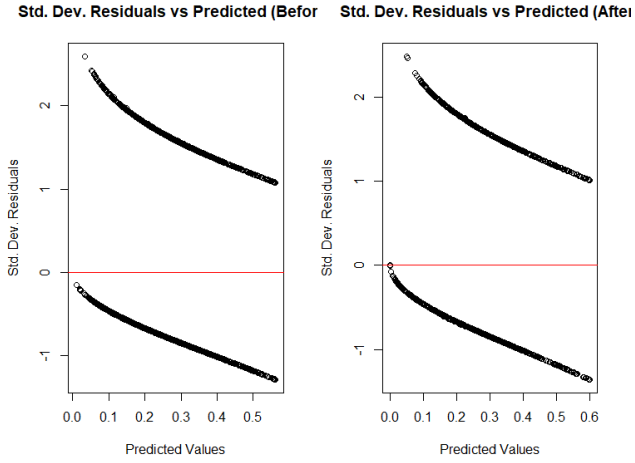


Figure 9: You can see there has been an improvement in the general structure

As a final measure, if we compare the AIC of this final model it works out to be 7767.2 which is an improvement to the model.

5 Interpretation

Note that in this GLM that we've chosen a Binomial with a logit canonical link, the way the variables map is such that:

$$\pi_i = \frac{e^{\eta_i}}{1 + e^{\eta_i}}$$

where η is the linear predictor for this GLM. And here, this corresponds to the final model:

$$\eta_i = \beta_0 + \beta_1 \times \text{aahisp} + \beta_2 \times \text{ban...}$$

and so the β are interpreted on the log-odds scale. Exponentiating them gives odds ratios which quantify the multiplicative effect of each predictor on the odds of smoking. More concretely, β_j gives the change in log odds of success when the corresponding explanatory variable x_j increases by 1 unit, which we estimate by β_j hat.

The results of the final model reveal several significant predictors of smoking behavior. The presence of a workplace smoking ban decreases the odds of smoking by approximately 22.8% (OR = 0.773, 95% CI: [0.701, 0.851], $p < 0.001$). Age is positively associated with smoking, with each additional year of age increasing the odds of smoking by 6.1% (OR = 1.061, 95% CI: [1.037, 1.086], $p < 0.001$). However, the quadratic term for age (*age2*) suggests a slight non-linear effect, with the odds ratio close to 1 (OR = 0.999, 95% CI: [0.999, 1.000], $p < 0.001$).

Education level has a strong inverse relationship with smoking. Compared to high school dropouts, high school graduates have 43.8% lower odds of smoking (OR = 0.562, 95% CI: [0.459, 0.689], $p < 0.001$), and individuals with some college education have 61.8% lower odds (OR = 0.382, 95% CI: [0.310, 0.472], $p < 0.001$). College graduates and those with a Master's degree or higher exhibit even greater reductions in smoking odds (OR = 0.168, 95% CI: [0.133, 0.213], and OR = 0.113, 95% CI: [0.084, 0.150], respectively, both $p < 0.001$).

Being female reduces the odds of smoking by 13.3% (OR = 0.867, 95% CI: [0.779, 0.964], $p = 0.0087$). For African-American or Hispanic individuals (*aahisp*), the interaction effects show complex relationships,

Variable	Odds Ratio (OR)	95% CI (Lower)	95% CI (Upper)	p-value
Intercept	0.454	0.277	0.738	0.0015
aahisp	0.357	0.247	0.514	¡0.001
ban	0.773	0.701	0.851	¡0.001
age	1.061	1.037	1.086	¡0.001
age2	1.000	0.999	1.000	¡0.001
education: HS Graduate	0.562	0.459	0.689	¡0.001
education: Some College	0.382	0.310	0.472	¡0.001
education: College Graduate	0.168	0.133	0.213	¡0.001
education: Master's+	0.113	0.084	0.150	¡0.001
female	0.867	0.779	0.964	0.0087
aahisp:age2	1.000	1.000	1.001	0.0047
aahisp:education: HS Graduate	1.506	1.062	2.134	0.0213
aahisp:education: Some College	1.838	1.278	2.640	¡0.001
aahisp:education: College Graduate	3.046	1.853	4.922	¡0.001
aahisp:education: Master's+	2.516	0.984	5.614	0.0353
aahisp:female	0.782	0.611	0.999	0.0491

Table 1: Odds Ratios, Confidence Intervals, and p-values for the Final Model

particularly with education. For example, among African-American or Hispanic individuals, having some college education increases the odds of smoking by 83.8% compared to high school dropouts (OR = 1.838, 95% CI: [1.278, 2.640], $p < 0.001$), and having a college degree triples the odds of smoking (OR = 3.046, 95% CI: [1.853, 4.922], $p < 0.001$). Similarly, African-American or Hispanic individuals with a college degree have more than triple the odds of smoking compared to those who dropped out of high school (OR = 3.046, 95% CI: [1.853, 4.922], $p < 0.001$). However, the interaction between *aahisp* and individuals with a Master's degree or higher results in a smaller but still significant increase in odds (OR = 2.516, 95% CI: [0.984, 5.614], $p = 0.035$).

The interaction between race/ethnicity and age (*aahisp:age2*) is also significant (OR = 1.000, 95% CI: [1.000, 1.001], $p = 0.0047$), suggesting a subtle non-linear modification in how age affects the odds of smoking for African-American or Hispanic individuals. Finally, the interaction between race/ethnicity and gender (*aahisp:female*) indicates that being female and African-American or Hispanic reduces the odds of smoking (OR = 0.782, 95% CI: [0.611, 0.999], $p = 0.049$), suggesting a protective effect for this subgroup.

In summary, these interaction terms highlight that the relationships between predictors and smoking behaviour are not uniform across all subgroups. Exactly what these interactions mean in the real world example are beyond the scope of this analysis - this is requires further psychophysiological and biological analysis to determine the meaning of these affects. The varying effects underscore the importance of accounting for these interactions to accurately model and interpret the odds of smoking.

In summary, the model demonstrates significant associations between workplace smoking bans, age, education, gender, and ethnicity, with strong protective effects of education and workplace bans against smoking. Confidence intervals provide robust evidence for these effects, excluding 1 in most cases, thereby reinforcing the statistical significance of the predictors.

6 Should Education be included as continuous instead of factors?

In the entire analysis we've treated the education variable as a categorical variable with factors. However, if we consider that it is a continuous variable this could result in a better model. To assess whether the education variable can be treated as a continuous variable instead of as a factor, two models were compared:

one with education as a factor and the other with education as a continuous numeric variable.

$$\text{AIC}_{\text{factors}} = 7767.25 \qquad \text{and} \qquad \text{AIC}_{\text{numeric}} = 7889.54$$

The AIC for the model with education as a factor is lower than the AIC for the continuous model which suggests that treating education as a factor results in a better fit and better captures the variability. Education has non-linear effects on the odds of smoking, which are better captured by treating it as a categorical variable. Treating education as continuous would oversimplify the relationship, leading to a poorer model fit.

Furthermore, a likelihood ratio test (LRT) was conducted to statistically compare the two models. The LRT yielded a test statistic of 18.85 with 4 degrees of freedom and a p-value of 0.001, indicating that the factor model provides a significantly better fit to the data compared to the continuous model.

Appendix

```
## Second Stats Practical
# Load necessary libraries
library(ggplot2)
library(dplyr)
library(car)
library(sandwich)
library(lmtest)# For diagnostics
# Load the data
smoke <- read.csv(file.choose())

# Exploratory Data Analysis
## Summary statistics
summary(smoke)
# Ensure education is treated as a factor
smoke$education <- factor(smoke$education, levels = 1:5,
                          labels = c("HS Dropout", "HS Graduate", "Some College", "
                              College Graduate", "Master's+"))
# Exploratory Data Analysis (EDA)
## Summary Statistics
summary_stats <- smoke %>%
  summarise(
    smoker_mean = mean(smoker),
    ban_mean = mean(ban),
    age_mean = mean(age),
    age_sd = sd(age),
    age2_mean = mean(age2),
    aahisp_mean = mean(aahisp),
    female_mean = mean(female)
  )
print(summary_stats)
## Distribution of Variables
### Age Distribution
ggplot(smoke, aes(x = age)) +
  geom_histogram(fill = "skyblue", bins = 30) +
  labs(title = "Age Distribution", x = "Age", y = "Frequency") +
  theme_minimal()
### Education Levels and Smoking Status
ggplot(smoke, aes(x = education, fill = factor(smoker))) +
  geom_bar(position = "dodge") +
  labs(title = "Education Levels and Smoking Status", x = "Education Level", y = "
      Count", fill = "Smoking Status") +
  theme_minimal()
### Education Levels
ggplot(smoke, aes(x = education)) +
```

```
  geom_bar(fill = "darkgreen") +
  labs(title = "Distribution of Education Levels", x = "Education Level", y = "
      Count") +
  theme_minimal()

## Bivariate Analysis
### Smoking vs Age
ggplot(smoke, aes(x = age, y = smoker)) +
  geom_jitter(alpha = 0.5, height = 0.05, width = 0) +
  labs(title = "Smoking Status vs Age", x = "Age", y = "Smoking Status") +
  theme_minimal()
### Smoking vs Education
ggplot(smoke, aes(x = education, fill = factor(smoker))) +
  geom_bar(position = "fill") +
  labs(title = "Smoking Status by Education Level", x = "Education Level", y = "
      Proportion", fill = "Smoking Status") +
  theme_minimal()
### Smoking vs Work Area Ban
ggplot(smoke, aes(x = factor(ban), fill = factor(smoker))) +
  geom_bar(position = "fill") +
  labs(title = "Smoking Status by Work Area Ban", x = "Work Area Ban", y = "
      Proportion", fill = "Smoking Status") +
  theme_minimal()
### Smoking vs Race/Ethnicity
ggplot(smoke, aes(x = factor(aahisp), fill = factor(smoker))) +
  geom_bar(position = "fill") +
  labs(title = "Smoking Status by Race/Ethnicity", x = "African-American or
      Hispanic", y = "Proportion", fill = "Smoking Status") +
  theme_minimal()
### Smoking vs Gender
ggplot(smoke, aes(x = factor(female), fill = factor(smoker))) +
  geom_bar(position = "fill") +
  labs(title = "Smoking Status by Gender", x = "Female", y = "Proportion", fill = "
      Smoking Status") +
  theme_minimal()
### Exploratory data analysis
# GLM Modelling
## Fit GLM with binomial canonical link
smoke_glm <- glm(smoker ~ (aahisp * (ban + age + age2 + education + female)),
                 family = binomial(link = "logit"), data = smoke)
summary(smoke_glm)
# Model Selection using Likelihood Ratio Tests (LRT)
## Start with Full Model
current_model <- smoke_glm
# Identify Candidates for LRT Based on p-values
coeff_summary <- summary(smoke_glm)$coefficients
candidates <- rownames(coeff_summary)[coeff_summary[, "Pr(>|z|)"] > 0.05 & !is.na(
    coeff_summary[, "Pr(>|z|)"])]
# Model Selection using Likelihood Ratio Tests (LRT)
current_model <- smoke_glm
for (term in candidates) {
  reduced_model <- update(current_model, paste(".~.-", term))
  lrt <- anova(reduced_model, current_model, test = "Chisq")

  print(paste("Testing term:", term))
  print(lrt)  # Show LRT results

  if (!is.na(lrt$`Pr(>Chi)`[2]) && lrt$`Pr(>Chi)`[2] > 0.05) {
    current_model <- reduced_model
  }
}
# Final Model
```

```
final_model <- current_model
summary(final_model)
# Compare AIC of Full and Final Models
full_model_aic <- AIC(smoke_glm)
final_model_aic <- AIC(final_model)
cat("AIC of Full Model:", full_model_aic, "\n")
cat("AIC of Final Model:", final_model_aic, "\n")
# Model Diagnostics
## Pearson and Deviance Residuals
residuals_pearson <- residuals(final_model, type = "pearson")
residuals_deviance <- residuals(final_model, type = "deviance")
par(mfrow = c(1, 2))
plot(residuals_pearson, main = "Pearson Residuals", ylab = "Residuals", xlab = "
    Index")
abline(h = 0, col = "red")
plot(residuals_deviance, main = "Deviance Residuals", ylab = "Residuals", xlab = "
    Index")
abline(h = 0, col = "red")
## Leverage and Standardized Residuals
influence <- influence.measures(final_model)
par(mfrow = c(1, 1))
plot(hatvalues(final_model), main = "Leverage", ylab = "Leverage", xlab = "Index")
abline(h = 2 * mean(hatvalues(final_model)), col = "blue", lty = 2)
## Cook's Distance
plot(cooks.distance(final_model), main = "Cook's Distance", ylab = "Cook's Distance
    ", xlab = "Index")
abline(h = 4 / nrow(smoke), col = "red", lty = 2)
## Estimation of Dispersion Parameter
dispersion <- sum(residuals_pearson^2) / final_model$df.residual
cat("Dispersion parameter estimate:", dispersion, "\n")
#Type 5
# Model Diagnostics
## Standardized Deviance Residuals
par(mfrow = c(2, 2))
standardized_residuals <- rstandard(final_model, type = "deviance")
# Plot Standardized Residuals vs Predicted Values
predicted_values <- predict(final_model, type = "response")
plot(predicted_values, standardized_residuals,
     main = "Standardized Deviance Residuals vs Predicted Values",
     xlab = "Predicted Values", ylab = "Standardized Deviance Residuals")
abline(h = 0, col = "red")
# Q-Q Plot for Standardized Deviance Residuals
qqnorm(standardized_residuals, main = "Q-Q Plot of Standardized Deviance Residuals"
    )
qqline(standardized_residuals, col = "red")

# Histogram of Standardized Deviance Residuals
hist(standardized_residuals, breaks = 20,
     main = "Histogram of Standardized Deviance Residuals",
     xlab = "Standardized Deviance Residuals", col = "skyblue", border = "white")
# Cook's Distance
plot(cooks.distance(final_model),
     main = "Cook's Distance",
     ylab = "Cook's Distance", xlab = "Index")
abline(h = 4 / nrow(smoke), col = "red", lty = 2)
# Remove Troublesome Points Based on Diagnostics
high_influence_points <- which(cooks.distance(final_model) > (4 / nrow(smoke)))
cat("High Influence Points:", high_influence_points, "\n")
# Refit Model Without High-Influence Points
smoke_clean <- smoke[-high_influence_points, ]
final_model_clean <- glm(smoker ~ ban + age + age2 + education + aahisp + female +
    aahisp:ban + aahisp:age + aahisp:education + aahisp:female,
```

```
                          family = binomial(link = "logit"), data = smoke_clean)
summary(final_model_clean)
# New Model Diagnostics
## Comparison Plots (Before and After)
par(mfrow = c(1, 2))
# Cook's Distance Before and After
plot(cooks.distance(final_model),
     main = "Cook's Distance (Before)",
     ylab = "Cook's Distance", xlab = "Index")
abline(h = 4 / nrow(smoke), col = "red", lty = 2)
plot(cooks.distance(final_model_clean),
     main = "Cook's Distance (After)",
     ylab = "Cook's Distance", xlab = "Index")
abline(h = 4 / nrow(smoke_clean), col = "red", lty = 2)
# Standardized Deviance Residuals Before and After
par(mfrow = c(1, 2))
standardized_residuals_clean <- rstandard(final_model_clean, type = "deviance")
plot(predicted_values, standardized_residuals,
     main = "Std. Dev. Residuals vs Predicted (Before)",
     xlab = "Predicted Values", ylab = "Std. Dev. Residuals")
abline(h = 0, col = "red")
plot(predict(final_model_clean, type = "response"), standardized_residuals_clean,
     main = "Std. Dev. Residuals vs Predicted (After)",
     xlab = "Predicted Values", ylab = "Std. Dev. Residuals")
abline(h = 0, col = "red")
# Q-Q Plots Before and After
par(mfrow = c(1, 2))
qqnorm(standardized_residuals, main = "Q-Q Plot (Before)")
qqline(standardized_residuals, col = "red")
qqnorm(standardized_residuals_clean, main = "Q-Q Plot (After)")
qqline(standardized_residuals_clean, col = "red")
# Leverage Before and After
par(mfrow = c(1, 2))
plot(hatvalues(final_model),
     main = "Leverage (Before)",
     ylab = "Leverage", xlab = "Index")
abline(h = 2 * mean(hatvalues(final_model)), col = "blue", lty = 2)
plot(hatvalues(final_model_clean),
     main = "Leverage (After)",
     ylab = "Leverage", xlab = "Index")
abline(h = 2 * mean(hatvalues(final_model_clean)), col = "blue", lty = 2)


# Calculate odds ratios and 95% confidence intervals
coefficients_summary <- summary(final_model)$coefficients
odds_ratios <- exp(coefficients_summary[, "Estimate"])
conf_int <- exp(confint(final_model))
# Combine results into a readable format
results <- data.frame(
  Variable = rownames(coefficients_summary),
  Odds_Ratio = odds_ratios,
  Lower_95CI = conf_int[, 1],
  Upper_95CI = conf_int[, 2],
  P_Value = coefficients_summary[, "Pr(>|z|)"]
)
print(results)
# Test whether education can be treated as continuous
## Refit model with education as continuous
smoke_clean$education_numeric <- as.numeric(smoke_clean$education)
education_continuous_glm <- glm(smoker ~ ban + age + age2 + education_numeric +
    aahisp + female + aahisp:ban + aahisp:age + aahisp:education_numeric + aahisp:
    female,
```

```
                                family = binomial(link = "logit"), data = smoke_
                                    clean)
summary(education_continuous_glm)
# Compare models using AIC and Likelihood Ratio Test
cat("AIC (Factor Education):", AIC(final_model_clean), "\n")
cat("AIC (Continuous Education):", AIC(education_continuous_glm), "\n")
anova(final_model_clean, education_continuous_glm, test = "Chisq")
```